



Audio2Tool

Speak, Call, Act — A Dataset for Benchmarking Speech Tool Use

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This talk in one slide

QUESTION

Can today's SpeechLMs turn a spoken request into a correct, executable tool call under realistic acoustic conditions?

WHAT WE BUILT

Audio2Tool. ~30K spoken queries with gold tool calls • 152 tools in 3 domains • 8 complexity tiers • 330 cloned voices • real in-car and indoor noise • open-weight baselines: end-to-end SpeechLMs vs. Whisper cascades.

WHAT WE FOUND



Tool selection nearly solved for direct commands

92% tool accuracy for the best model on Tier 1



Argument grounding is the bottleneck (not ASR)

exact match $\leq 16\%$, even with gold transcripts



Composition, implicit intent, competing speakers

still open: 33–75% tool accuracy on Tiers 3–4 and 7–8



End-to-end does not yet beat a strong cascade

SpeechLMs pick tools better; text LLMs fill arguments better

The rest of the talk is the evidence.



Voice assistants are becoming agents: the output is now an action

THEN • classify the intent



NOW • emit the executable call



Stricter correctness than SLU. A tool call must be schema-valid, fully specified, and correctly ordered. A semantically plausible output with one wrong argument is not "almost right"; it is the wrong action.



Speech adds failure modes that text benchmarks never see



Phonetic ambiguity

"Fifteen" vs. "fifty", "seventy-two" vs. "seventy". Arguments are where sound-alikes bite.

*"Set the fan to fifteen" →
setCabinFanLevel(50)?*



In-the-wild acoustics

Engine, road, wind, HVAC, rain, turn signals, babble. Signal-to-noise can go negative.

-5 dB SNR on the highway



Self-repair

People change their mind mid-sentence. The final state, not the first mention, is the intent.

"Set an alarm for 7... wait, make it 8."



Competing speakers

Background speech carries valid commands. Only the primary speaker should be obeyed.

*Driver: "Navigate home."
Radio: "Set temperature to 72."*

Errors here do not produce a wrong label. They produce a wrong action: the wrong temperature, the wrong door, the wrong destination.



Existing benchmarks each cover part of the problem

	Speech input	Executable tool calls	Multi-intent & multi-turn	Accent & noise diversity	Diagnostic complexity tiers
Text function calling BFCL, ComplexFuncBench	No	Yes	Yes	No	Limited
Spoken language understanding SLURP, STOP, MAC-SLU	Yes	No (intents + slots)	Limited	Limited	No
Audio tool use BFCL-Audio, VoiceAgentBench, MFCL	Yes	Yes	Limited	Limited	No
Audio2Tool this work	Yes	Yes	Yes (8 tiers)	Yes (330 voices, noise)	Yes

Complementary, not competing. Audio2Tool keeps the executable-call formalism of BFCL, but makes the input spoken, acoustically diverse, and organised so each tier isolates a distinct failure mode.



Audio2Tool at a glance

~30K

spoken queries, each with a
gold tool call

152

verified tools grounded in real
APIs

23

functional categories

3

domains: smart car · smart
home · wearables

8

complexity tiers, one failure
mode each

330

cloned voices from 4 corpora,
4 regions

2

zero-shot voice-cloning TTS
engines

8+

noise sources: engine, road,
wind, HVAC, rain...

Dataset, benchmark code and audio samples: audio2tool.github.io



Taxonomy grounded in real APIs: 3 domains, 23 categories, 152 tools



Smart Car

14 categories · climate, driving dynamics, charging, body control, navigation, maintenance...

```
setZoneTemperature(zone, temp)
openFrontTrunk()
```



Smart Home

lighting, appliances, automation sensors, security, media...

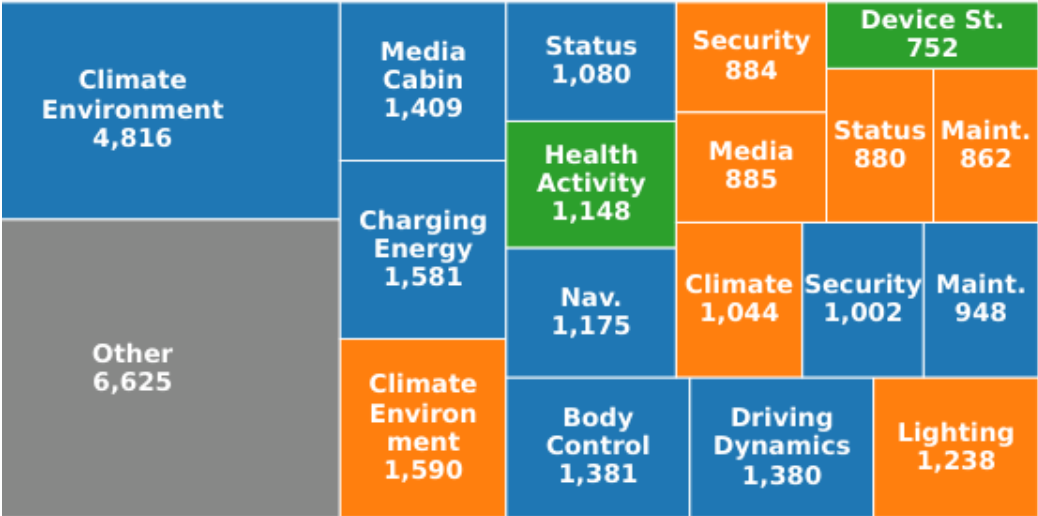
```
setLightState(device, state)
armSecuritySystem(mode)
```



Wearables

health & activity tracking, device state and connectivity

```
startWorkout(activity)
setDoNotDisturb(enabled, mode)
```



Smart Car Smart Home Wearables

Query share by category (paper Fig. 2c)

Operational intent state-altering ("set temperature") vs. monitoring ("check battery")

Domain specificity device-specific categories kept (driving dynamics, activity tracking)



Eight tiers, eight failure modes

1 Direct

Short command, no parameters
(2–6 words)

"Open the trunk."

2 Parametric

Explicit slot values to extract (3–12
words)

"Set the temperature to 72."

3 Multi-intent

2–3 tool calls in one utterance, in
order

*"Defrost the windshield and find a
charger."*

4 Implicit

State or complaint; the tool is only
implied

"I'm freezing."

5 Needle in a haystack

Intent buried in 25–60 words of
tangent

*"...anyway, long day, could you... oh,
lock the doors."*

6 Correction

Mid-utterance repair; the final
state wins

*"Set an alarm for 7... wait, make it
8."*

7 Conversation

Multi-turn USER/AGENT dialogue,
20–200 words, persistent state

*"...and the same for the passenger
seat."*

8 Intent blending

Background speaker carries a
second, valid command

*Driver: "Navigate home."
Radio: "Set temperature to 72."*

● Basics (T1–T2) ● Composition & inference (T3–T4) ● Realism: long context, repair, dialogue, competing speakers (T5–T8)



From taxonomy to ~30K verified queries



1. Tier-specific prompting

GPT-5.2 · Gemini 2.5 Pro · Claude Opus generate queries over the 152-tool taxonomy, with gold tool call(s) and arguments



2. LLM-as-judge (disjoint models)

GPT-5.1 · Gemini 2.5 Pro score correctness, difficulty and variability; never the same models that generated



3. Manual verification

Every query a judge failed was checked by hand, together with its ground-truth tools



4. Gold labels per tier

tool call(s) + arguments, plus correction type (T6), reasoning (T4/T5/T7), primary-speaker calls (T8)

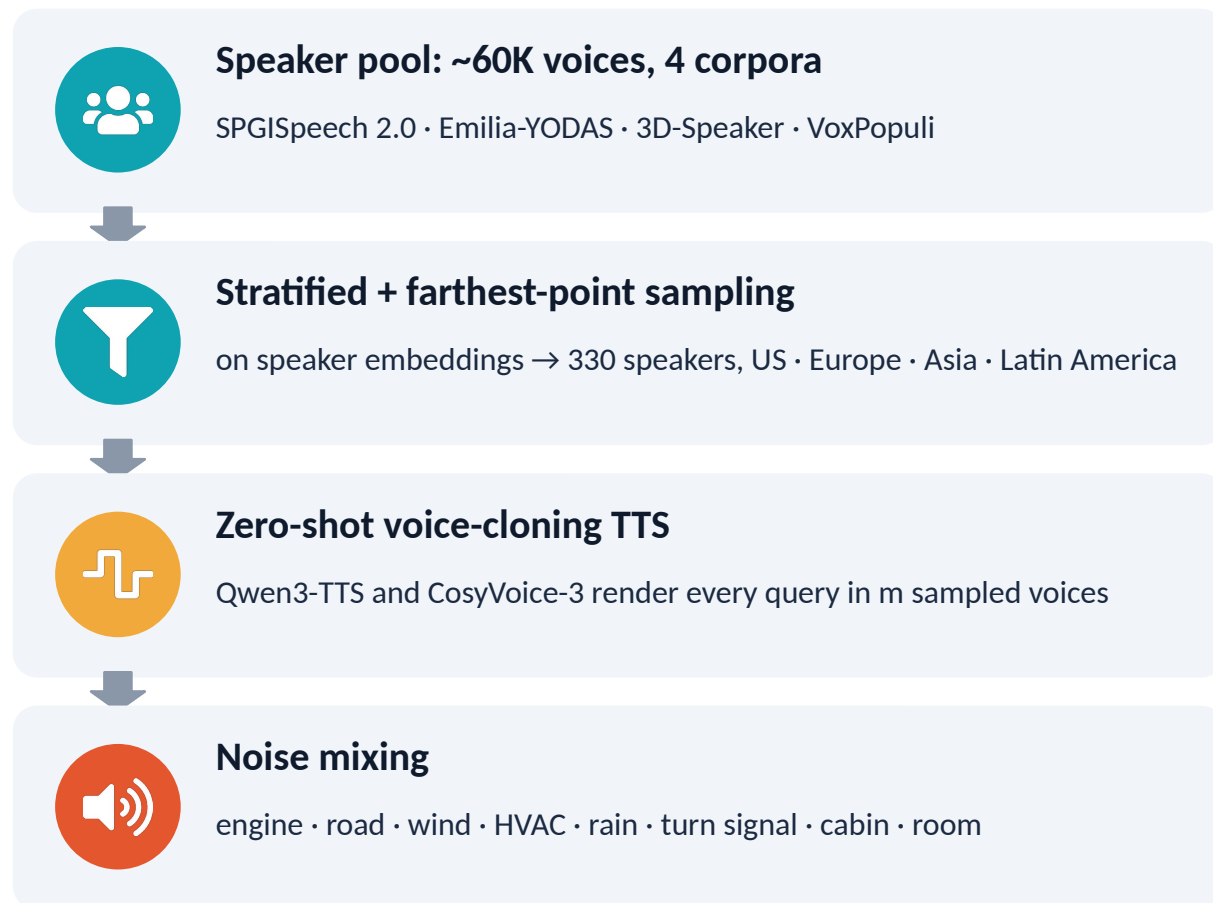
Queries per tier



Tier 3 composition: 2 calls 61.8% · 3 calls 36.2% · 4–5 calls 1.9%
Smart Car is the majority domain in every tier (paper Fig. 2a).



Making it sound real: 330 cloned voices, real noise



Speaker corpora (paper Table 2)

Corpus	Speakers	Regions	Selected
SPGISpeech 2.0	41,593	US, Asia, Latin America	100
Emilia-YODAS	7,092	US, China	100
3D-Speaker	10,000	China	30
VoxPopuli	1,180	US, Europe	100
Total	59,865	4 regions	330

Algorithm 1, in one line

```
S ← FPS({f(s)}, K)
for q in Q: for s in sample(S, m):
     $\tilde{a} \leftarrow \text{Mix}(\text{TTS}(q, s), n), \quad n \sim N$ 
```

Every query is heard in several voices and acoustic conditions.



Evaluation protocol: two system families, three metrics



End-to-end SpeechLMs • audio → tool call

Qwen3-Omni-30B • Qwen2.5-Omni-7B • Kimi-Audio-7B
Step-Audio-2-7B • Audio-Flamingo-3-8B

Open weights only • 7B to 30B • zero-shot with tool definitions in the prompt



Cascaded • audio → Whisper-v3 → text LLM

Qwen3 1.7B / 4B / 8B • Gemma-3 12B / 27B

+ Oracle transcript: gold text → same LLMs. Upper bound that isolates ASR damage.

Tier 8 (intent blending) is audio-only and is not scored for text/oracle systems.

Three metrics, one worked example

```
gold  setZoneTemperature(zone="Driver", temperature=72)
pred  setZoneTemperature(zone="Driver", temperature=27)
```

1

Tool Accuracy

Right tool name(s), in the right order

0

Exact Match

Whole call identical after normalisation

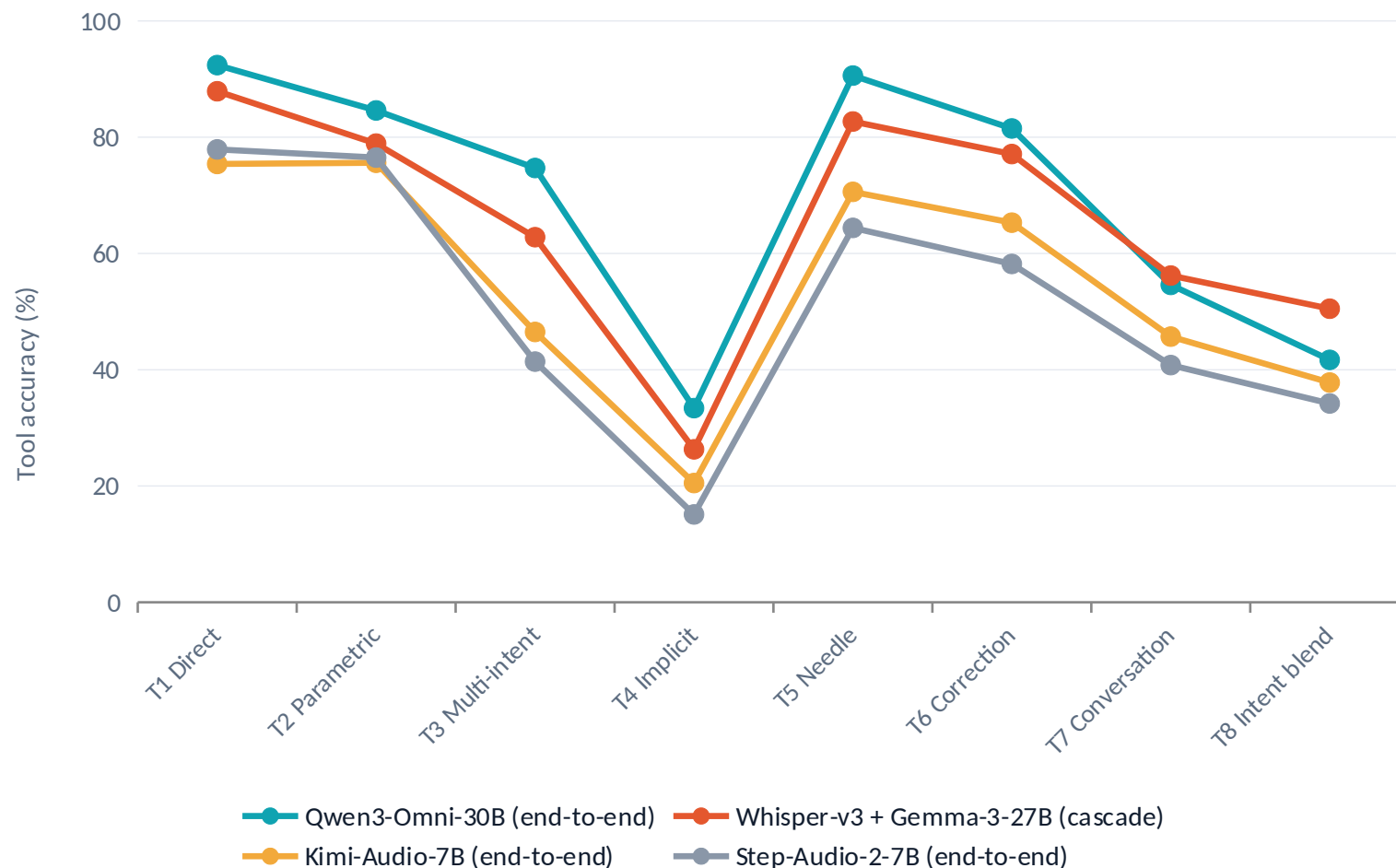
0.5

Parameter F1

Micro-averaged over arguments (1 of 2 correct)



Tool accuracy collapses when the query needs composition or inference



92% → 33%

best model, direct command (T1) → implicit intent (T4)

T5 ≈ 91%

long, rambling input alone is not the problem

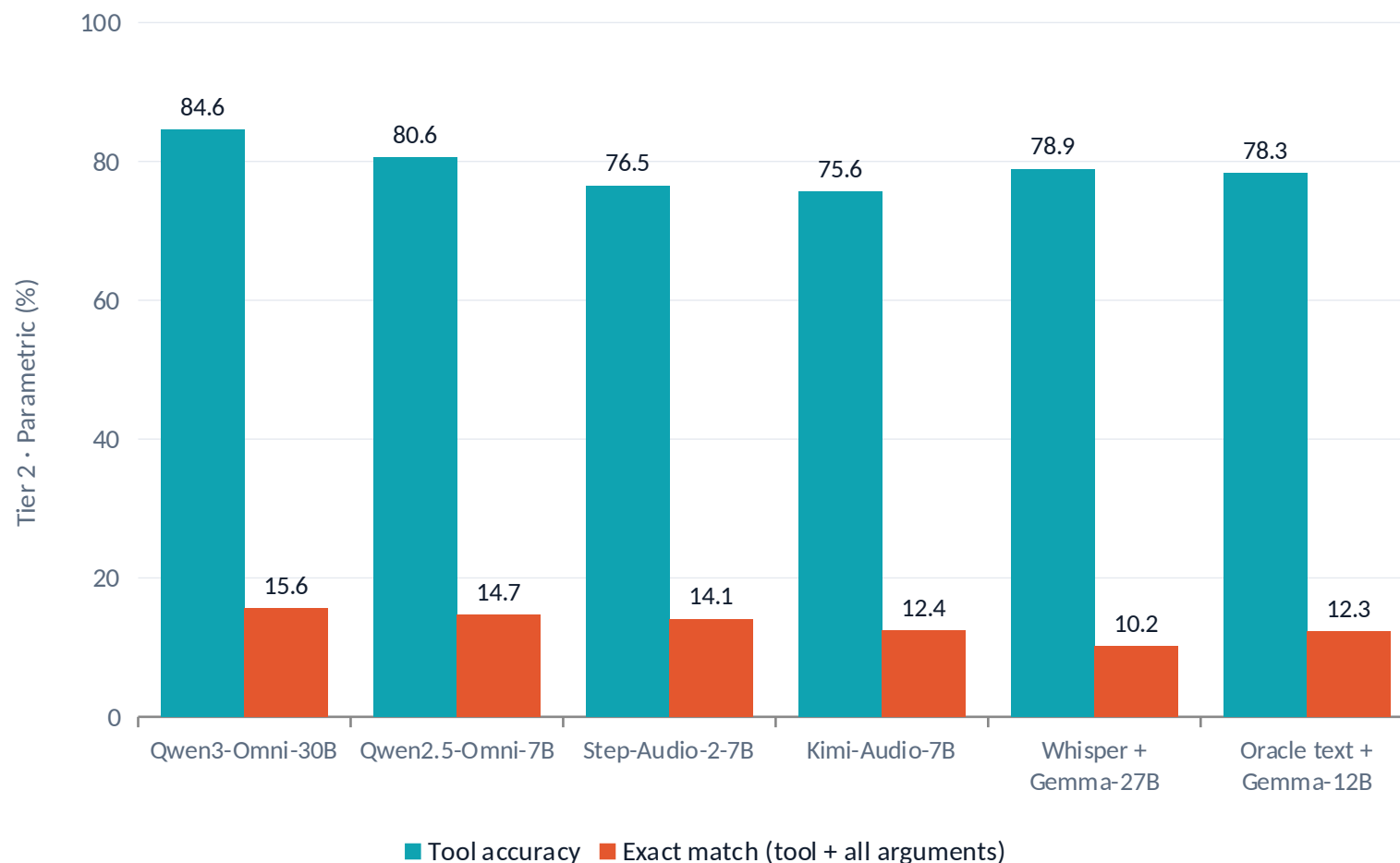
≤ 56%

everyone, on conversation (T7) and intent blending (T8)

Length is cheap; reasoning is expensive.



Right tool, wrong arguments: exact match $\leq 16\%$ even on plain parametric commands



~80%

pick the right tool

10–16%

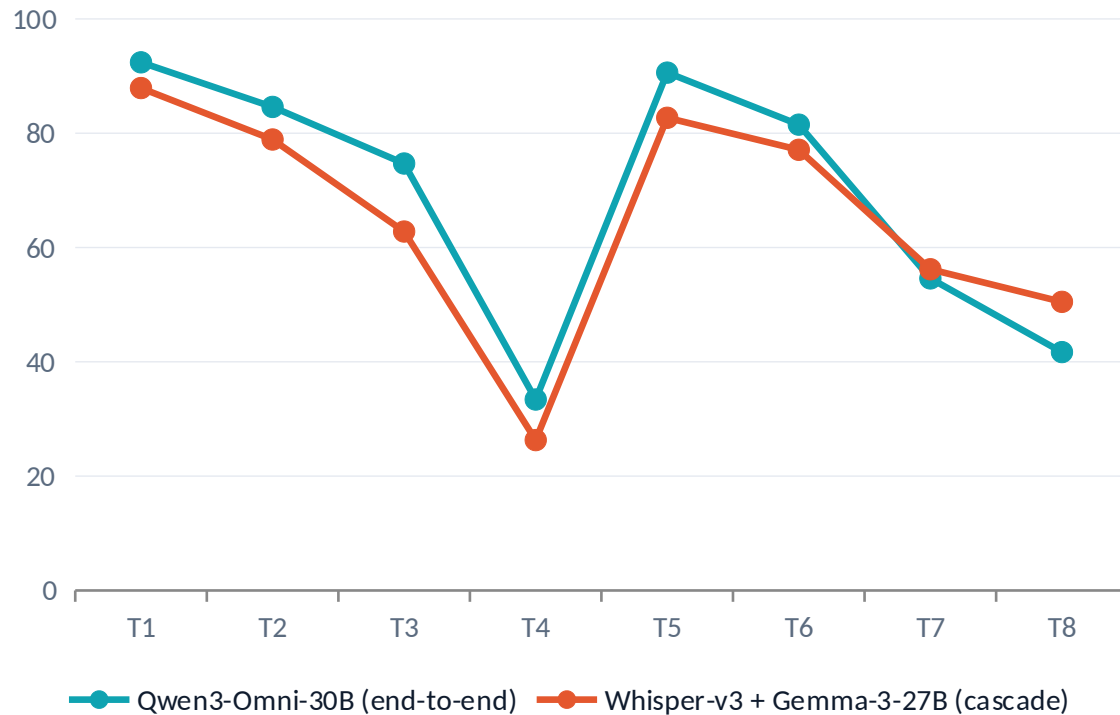
get every argument right

Not an ASR problem. With a perfect transcript (oracle), Gemma-3-12B still reaches only 12.3% exact match. The bottleneck is argument grounding: units, ranges, enumerations, defaults.

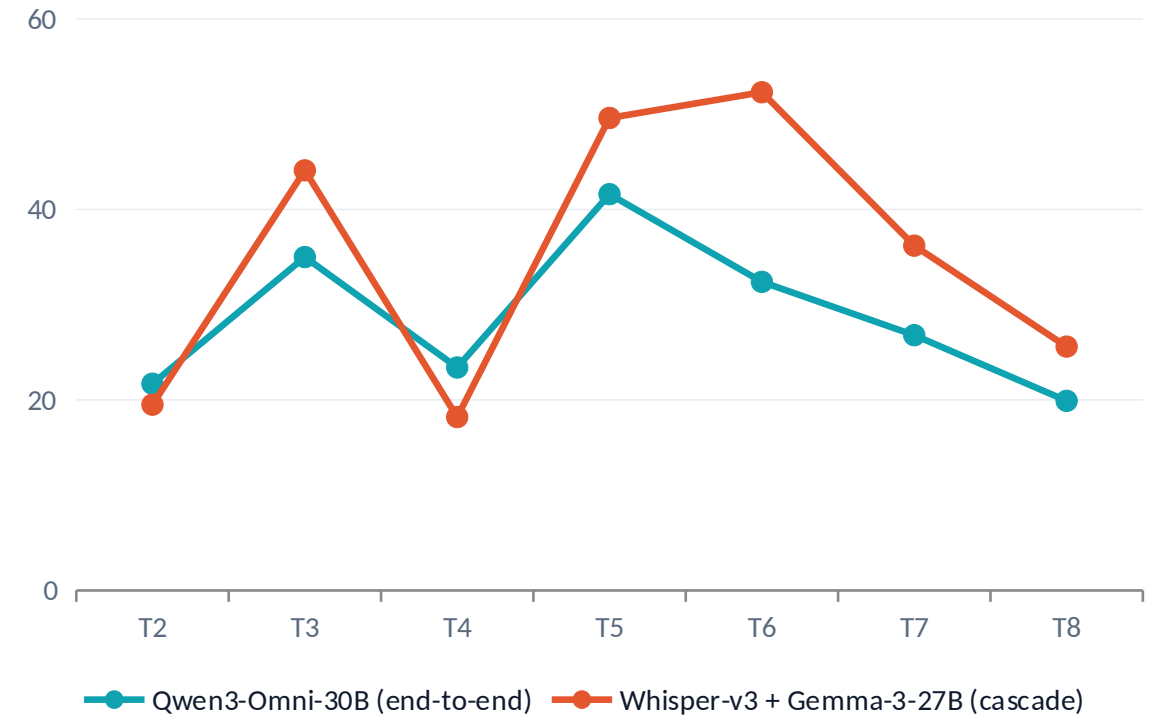


End-to-end SpeechLMs do not yet beat a strong cascade

Tool accuracy (%) • end-to-end wins 6 of 8 tiers



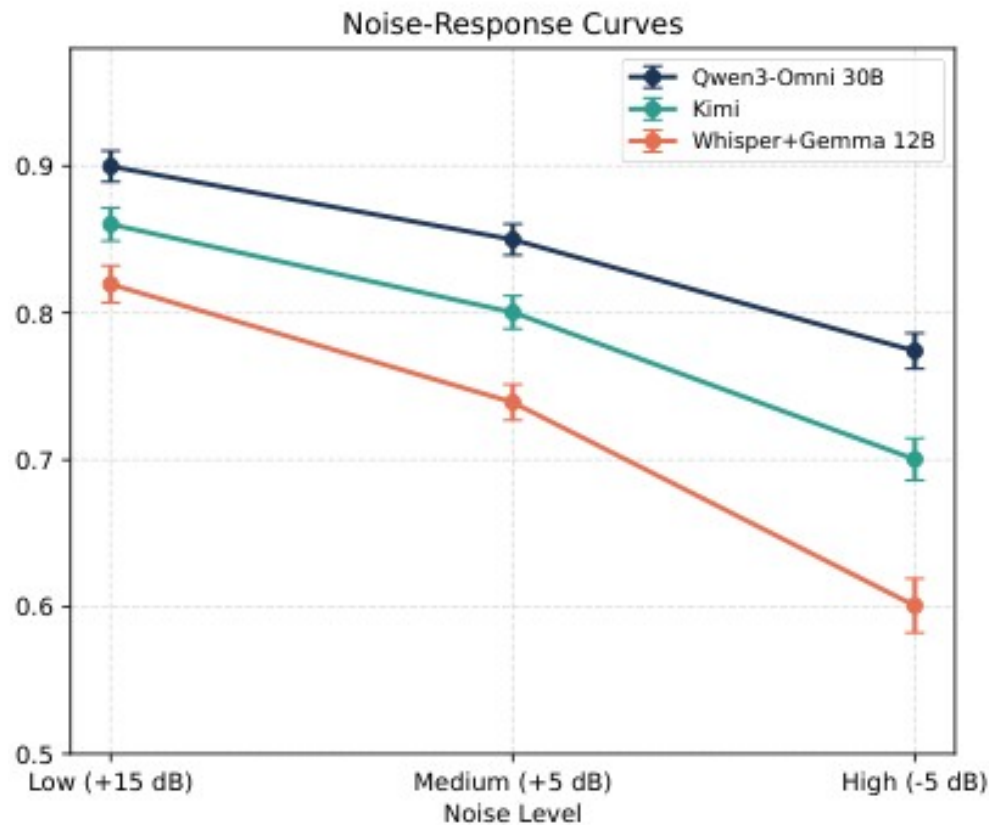
Parameter F1 (%) • cascade wins 5 of 7 tiers



Audio-native models hear the intent; text models still fill the arguments better. And ASR is not free: gold transcripts add 10–14 points of tool accuracy to the cascade (backup).



Noise degrades every system, and the cascade degrades fastest



Intent-classification F1 vs. noise level (paper Fig. 3, right). Error bars: std. over noise types.



3 noise families × 3 SNR levels

Babble, mechanical hum, impulsive (MS-SNSD) at +15, +5 and -5 dB



Monotonic degradation for everyone

F1 falls steadily from low to high noise for every system tested



Cascades compound ASR errors

Whisper + Gemma-3-12B loses the most; Qwen3-Omni-30B is the most robust



So noise is in the benchmark

Multiple noise types and levels are part of the released data, not just this ablation



What Audio2Tool tells us about speech tool use today



Tool selection is nearly solved for direct commands

>90% for the best model on Tier 1



Argument grounding is the bottleneck (not ASR)

Exact match $\leq 16\%$ on parametric commands, even with gold transcripts



Composition, implicit intent, competing speakers: open

Tiers 3–4 and 7–8 sit at 33–75% tool accuracy; long input alone is fine



End-to-end does not yet beat a strong cascade

SpeechLMs pick tools better; text LLMs fill arguments better; noise hurts cascades most

Next



Real recordings

beyond zero-shot cloned speech



Safety-critical scenarios

where a wrong action has real cost



Broader domains, languages

and per-accent, per-noise breakdowns



Use the tiers

report where a model fails, not just its average



Speak. Call. Act.

Dataset, evaluation harness and audio samples are released.
We would love to see your model on the leaderboard, one tier at a time.



audio2tool.github.io samples, dataset, benchmark



[arXiv:2604.22821](https://arxiv.org/abs/2604.22821) paper



github.com/RamitPahwa/Audio2Tool code



ramitpahwa@rivianvw.tech contact

Thank you • Questions?



audio2tool.github.io

listen to Tier 6, 7 and 8 samples



Full results by tier (paper Table 3)

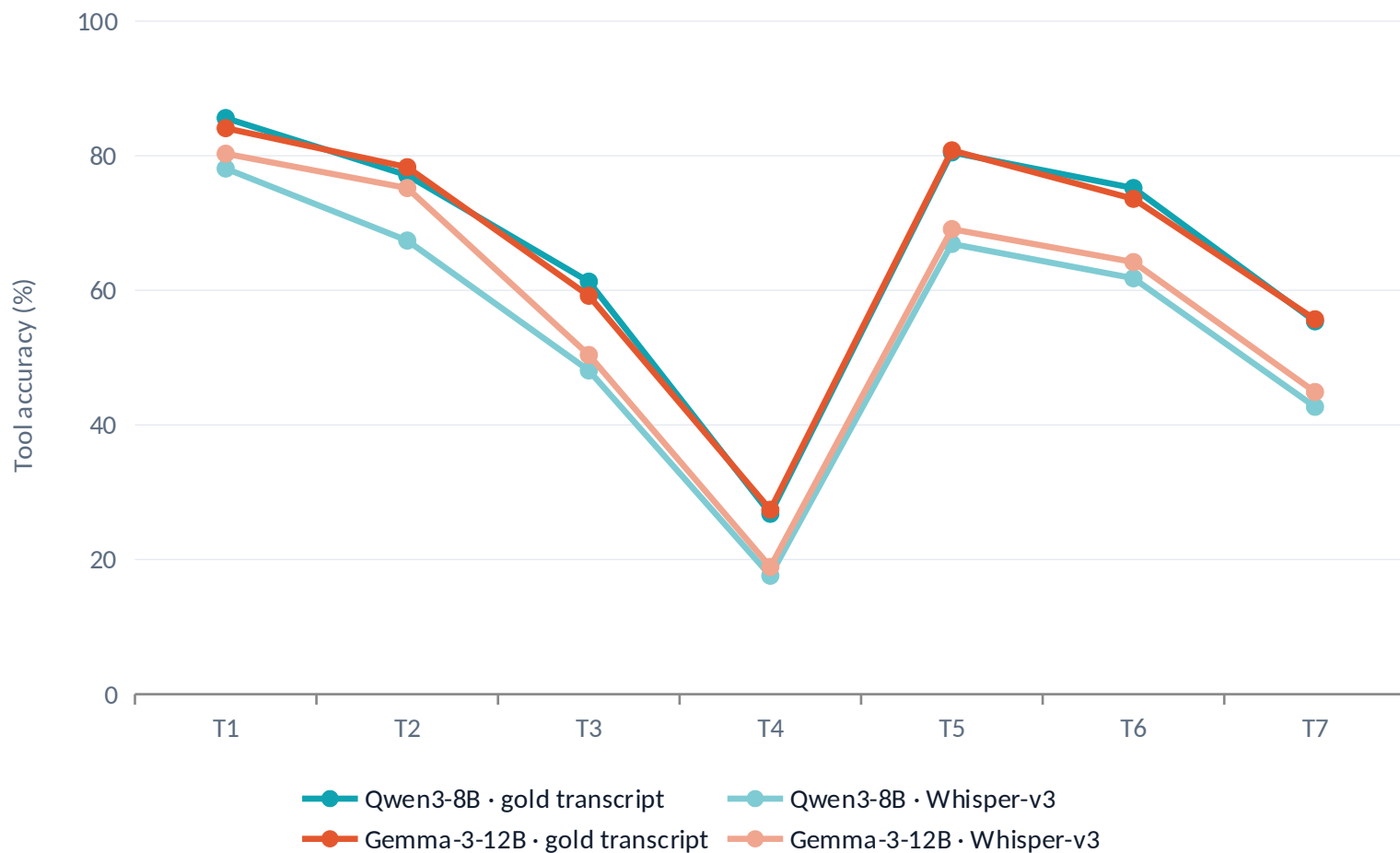
Table 3: Tool-calling performance across complexity tiers. Acc, EM, and F1 denote tool accuracy, exact match, and slot/parameter F1 scores respectively. F1 is undefined for Tier 1 queries, as no parameters exist.

Model	Tier-1			Tier-2			Tier-3			Tier-4			Tier-5			Tier-6			Tier-7			Tier-8		
	Acc	EM	F1	Acc	EM	F1	Acc	EM	F1	Acc	EM	F1	Acc	EM	F1	Acc	EM	F1	Acc	EM	F1	Acc	EM	F1
Qwen 1.7B	74.6	74.6	–	35.7	3.2	5.0	48.2	30.4	36.7	18.2	8.9	14.9	68.4	35.6	41.9	62.3	36.8	44.5	43.1	20.2	29.8	-	-	-
Qwen 4B	80.2	80.2	–	41.6	6.7	13.1	54.6	34.1	39.2	21.7	10.1	16.2	73.9	39.8	46.2	69.1	41.5	49.1	48.9	24.3	33.1	-	-	-
Qwen 8B	85.6	85.6	–	77.1	10.1	19.3	61.3	38.7	43.6	26.8	12.6	18.9	80.5	44.9	51.3	75.2	46.8	54.6	55.4	28.7	37.9	-	-	-
Gemma 12B	84.1	84.1	–	78.3	12.3	20.4	59.2	37.6	42.7	27.4	14.5	19.1	80.8	45	51.5	73.6	45.9	53.8	55.7	29.9	38.8	-	-	-
whisperv3 + Qwen 1.7B	68.2	68.2	–	33.6	2.9	4.2	37.6	22.4	29.8	11.4	5.6	10.8	56.9	27.8	33.2	51.4	25.9	31.4	33.8	14.7	23.4	18.9	6.0	18.9
whisperv3 + Qwen 4B	72.9	72.9	–	39.6	5.6	11.6	42.9	26.1	32.7	14.3	6.8	12.6	61.8	31.4	37.9	56.7	30.1	38.3	38.6	17.2	26.6	31.6	10.7	17.6
whisperv3 + Qwen 8B	78.1	78.1	–	67.4	10.8	15.4	48.1	29.4	35.1	17.6	8.2	14.1	66.9	34.1	41.9	61.8	34.5	44.2	42.7	20.1	30.1	43.5	11.9	19.5
whisperv3 + Gemma 12B	80.3	80.3	–	75.2	9.7	17.9	50.4	30.8	36.9	18.9	8.9	14.7	69.1	35.9	43.1	64.2	36.3	45.3	44.9	21.6	31.2	46.5	13.9	22.0
whisperv3 + Gemma 27B	87.9	87.9	–	78.9	10.2	19.5	62.8	36.9	44.1	26.3	11.9	18.2	82.7	42.3	49.6	77.1	44.1	52.3	56.2	26.4	36.2	50.5	15.7	25.6
Kimi-7B	75.4	75.4	–	75.6	12.4	15.8	46.5	30.4	32.1	20.5	9.0	12.3	70.6	38.6	43.4	65.3	39.1	45.1	45.7	24.5	31.3	37.8	10.6	16.7
Step-Audio2-7B	77.9	77.9	–	76.5	14.1	17.3	41.4	26.4	34.1	15.1	7.3	14.4	64.4	35.4	43.8	58.2	33.4	43.6	40.8	18.6	28.2	34.2	10.4	15.6
Audio-Flamingo-8B	64.5	64.5	–	75.3	12.4	15.7	40.6	25.4	32.3	7.9	4.6	2.3	58.9	28.5	31.6	44.6	21.6	31.3	34.5	15.7	24.5	30.4	8.1	14.5

Rows 1–4: oracle (gold transcript) + text LLM · rows 5–9: Whisper-v3 + text LLM · rows 10–14: end-to-end SpeechLMs. Acc = tool accuracy, EM = exact match, F1 = parameter F1. F1 undefined for Tier 1 (no parameters).



The ASR tax: gold transcripts add 10–14 points of tool accuracy to the same LLM



Qwen3-8B gap (gold - Whisper)

T1	+7.5
T2	+9.7
T3	+13.2
T4	+9.2
T5	+13.6
T6	+13.4
T7	+12.7

points of tool accuracy recovered with perfect ASR



Audio2Tool vs. SLU corpora (paper Table 1)

Dataset	Domains	Intents / tools	Slots / parameters	Multi-intent	Output
ATIS	1	16	41	No	intent + slots
SNIPS	2	7	4	No	intent + slots
FSC	2	6	2	No	intent + slots
SLURP	18	46	56	No	intent + slots
MixATIS	1	16	41	Yes	intent + slots
MixSNIPS	2	7	4	Yes	intent + slots
MAC-SLU	8	81	192	Yes	intent + slots
Audio2Tool	3	152	22	Yes	executable tool call(s)

Slot count is lower by construction: tool arguments are typed parameters with constrained ranges, not free-text spans.



Tier generation details: word budgets, prompt inputs and label fields (paper Fig. 1)

Tier 1: Direct Short, direct commands 2–6 words No parameter values Domain-specific guidance (car / home / wearables) query + tool_call	Tier 2: Parametric Commands with explicit values 3–12 words Explicit parameter values Filters no-param tools TTS-friendly numbers + extracted_params	Tier 3: Multi-Intent Combines 2–3 tool calls 4–18 words Uses: and / then / also Input: companion tools string (tools from same domain) + additional_tool_calls	Tier 4: Implicit Indirect, state-based commands 3–12 words State/complaint phrasing Input: arg_defaults + arg_constraints + reasoning	Tier 5: Needle in Haystack Long, rambling speech 25–60 words Filler words allowed Intent buried in tangent Run-on sentence structure + reasoning	Tier 6: Correction Mid-utterance repair 5–20 words guides: no wait / actually Input: related tools string (tools from same category) + original_tool_call + correction_type	Tier 7: Conversation Multi-turn dialogue 20–200 words 2–4 conversation turns USER: / AGENT: format + reasoning	Tier 8: Intent Blending Multi-speaker speech overlapping commands 4–18 words Multi-intent request Noisy audio with background speech + additional_tool_calls for primary speaker
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Generators GPT-5.2 · Gemini 2.5 Pro · Claude Opus

Judges (disjoint) GPT-5.1 · Gemini 2.5 Pro: correctness, difficulty, variability

Verification manual check of every judge-flagged query and its gold tools

TTS Qwen3-TTS · CosyVoice-3, zero-shot voice cloning; noise mixed per Algorithm 1